

TrES-3b

R-0432

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_230805 · created 2026-07-19T10:59:39+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460161.7132 +/- 0.0017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.194 +/- 0.036
Transit depth (Rp/Rs)^2	3.76 +/- 1.38 [%]
Orbital Inclination (inc)	81.13 +/- 1.2
Airmass coefficient 1 (a1)	1.103 +/- 0.013
Airmass coefficient 2 (a2)	-0.094 +/- 0.051
Scatter in the residuals of the lightcurve fit is	1.21 %
Best Comparison Star	None
Optimal Aperture	4.35
Optimal Annulus	17.22
Transit Duration (day)	0.048 +/- 0.017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.194 ± 0.036 vs 0.16309 (deviation 0.79σ)
Duration fit vs published	0.048 d vs 0.0567 d (deviation 0.47σ)
Mid-time O–C	-1.1 min
Depth SNR	5.0
Residual scatter	1.21 %

Light curve

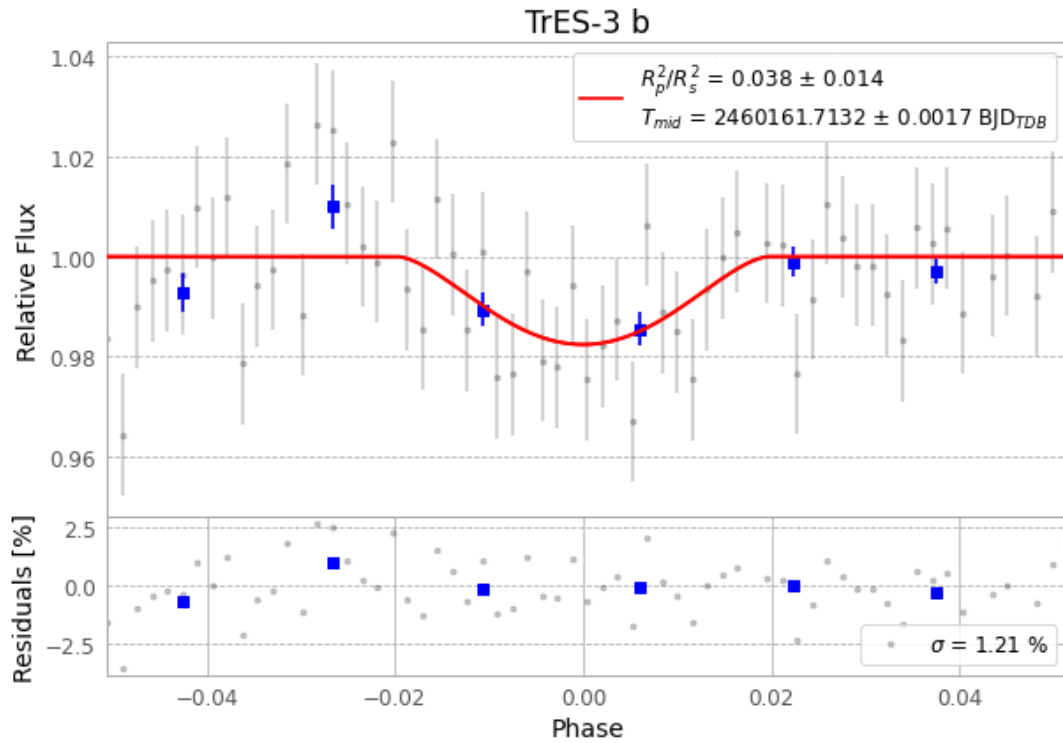


Figure 1 — FinalLightCurve_TrES-3 b_2023-08-04.png (SHA-256 45dfa0b03c3bf2cf...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [240, 198], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 4a483edaf4333a70...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

63 FITS frames (41 MB), TRES-3230805032715.FITS → TRES-3230805063920.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-230805.log (ca3b32c13b60...), FinalLightCurve_TrES-3 b_2023-08-04.png (45dfa0b03c3b...), FinalLightCurve_TrES-3 b_2023-08-04.csv (016e31607bf5...)

Compute resources

Wall time 200.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 10:59Z.